Yagati Prem Sagar

ee23btech11065@iith.ac.in | sagaryagati417@gmail.com | +91 9182646579 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.79
SRI CHAITANYA JUNIOR COLLEGE	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	97.00%
SRI CHAITANYA HIGH SCHOOL	Class X (Board of Secondary Education, Andhra Pradesh)	2021	100.00%

WORK EXPERIENCE

AGC Networks India Pvt. Ltd. | Data Scientist Intern

- **ML Forward & Inverse Modeling:** Trained KNN and Random Forest forward models and a GPR-based inverse model to predict slot-loaded patch geometry from target frequencies, using GPR to quantify uncertainty in the non-unique geometry-to-frequency mapping.
- **Differentiable Antenna Simulator:**Identified data limitations and built a physics based differentiable simulator in PyTorch, combining RLC impedance and inset feed theory for gradient based optimization over 10 design variables; validated geometries across multiple target frequencies.
- **Optimization Analysis:** Analysed optimization failures by separating infeasible targets from cases where design variables hit their bounds, identifying the source of target mismatch.
- **Optimization Dashboard:** Integrated the antenna optimization engine with a FastAPI based web interface for running optimization and displaying results.

PROJECTS

Perceptually Aware VDM's for Text to Video Generation| *Project under Prof . Sumohana Channappayya*

- Fine-tuned a VAE with an LGN inspired perceptual loss to make latent frame trajectories more consistent, then built a text conditioned VideoLDM using frozen Stable Diffusion spatial weights and trained new temporal layers to learn motion.
- Built an inference pipeline(VideoLDM-> SEINE-> Real-ESRGAN), extending 8 frame 256×256 clips to 24 frame, 1024×1024 temporally consistent videos.
- Benchmarked the LGN guided model against baseline on UCF 101 using FID/FVD, showing consistent improvement at each stage of pipeline.

Exploring the Design Space of Adaptive Noise Detection in Zero Shot-Noice Test Time Adaption|*CV Course Project*

- Implemented and analyzed the AdaND baseline with a frozen CLIP classifier and online trained linear noise detector using MCM score thresholding, then evaluated 11 variants across thresholding, detector architecture, training objective, and inference on CIFAR 10/100 paired with SVHN/LSUN/Places.
- Found that GMM thresholding struggles on CIFAR 100 due to overlapping score distributions, while confidence-margin training improved robustness by excluding ambiguous near-boundary pseudo labels.
- Found that detector improvements were limited by the quality of frozen zero shot CLIP pseudo labels, highlighting a pseudo label quality ceiling in the adaptation pipeline.

Boosting Lightweight Models for Mental Manipulation Detection | *NLP Course Project*

- Designed EvoSA, an evolutionary augmentation method that recombines same label dialogues by speech act and varies tone and wording while preserving labels, helping expand scarce manipulation data without manual re annotation.
- Developed a phased distillation approach combining rationale critique, rationale generation, and binary judgment, gradually shifting toward the core classification task to reduce conflicting gradients.
- Enabled a 3B parameter student to match or outperform much larger zero shot LLMs on manipulation detection, and extended the framework to technique and vulnerability classification.

Self Correcting Data Analysis Agent

- Built a LangGraph agent that answers natural language questions about CSV data by grounding an LLM in a schema summary of the dataset, generating pandas code, and running it in a separate subprocess with a timeout after a safety check that blocks unsafe imports and file operations.
- Added a self correction loop that sends execution errors and empty or NaN results back to the LLM along with previous failed attempts, with a retry limit and early stopping when the same error repeats.
- Evaluated the agent against hand verified answers, logging retries and latency per run, and extended it with LLM driven tool calling to compare a model chosen tool loop with the fixed pipeline on accuracy and latency.

SKILLS

Programming Languages: C, C++, Python, My SQL

Libraries/Frame Works: Numpy,Pandas,Tensor Flow,PyTorch,OpenCv,Matplotlib, LangChain ,Lang Graph

Others: Git/GitHub,Latex,VS code,Linux

RELEVANT COURSES

CS: Introduction to Programming,DBMS,Data Structures and Algorithms

AI/ML: Pattern Recognition and Machine Learning,Deep Learning,Natural Language Processing,Computer Vision

Mathematics: Matrix Theory ,Probability and Random Process,Differential Equations,Calculus-I & II,Vector Calculus,Complex Variables,Convex Optimization

Communication: Signals and Systems, Digital Signal Processing,Communication Systems,Wireless Communication

EXTRACURRICULAR

- Volunteered at the NSS Vidhyadhan event to teach school students
- Participated in Swachh Bharat cleanliness event, supporting community hygiene and public awareness.